Safe modal agents that coordinate in the Stag Hunt: size 13 is the minimum

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1 Introduction

Setting

In the Stag Hunt two players each choose stag or hare: both hunting stag pays most, hunting stag alone pays least, and hare is a safe middle option. Here the players are programs that read each other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL: p means "the opponent hunts stag against me" and q "the opponent hunts stag against HareBot", the agent that always hunts hare. Call an agent *safe* if no modal opponent ever ends up hunting hare while the agent hunts stag (the Stag Hunt reading of unexploitability). Two simple probes test willingness to coordinate: $E_1 = \neg\Box\bot$, which hunts stag unconditionally in practice, and $N_1 = \Box(\Box\bot \vee \neg q)$, which hunts stag against agents that do not hunt stag against HareBot. We take guarded formulas built from p, q, $\bot$, $\neg$, $\wedge$, $\vee$, $\Box$ and measure size as the number of nodes.

Result

- **No small safe coordinator.** Among the 56,503 agent classes of size at most 12, exactly 356 reach stag/stag with themselves, with E_1 and with N_1, and every one of them is exploited by some opponent of size at most 10. So no safe agent of size at most 12 has all three coordination properties.
- **Size 13 is enough.** The agent $X_1 = \Box(p \vee \Box p) \wedge (\Box\bot \vee \neg\Box\neg p)$ of size 13 is safe against every modal opponent and reaches stag/stag with itself, with FairBot, with E_1 and with N_1. More generally, the family $X_k = \Box(p \vee \Box^k p) \wedge (\Box\bot \vee \neg\Box\neg p)$ is safe for every $k \geq 1$, coordinates with itself and with FairBot, with E_j exactly when $j \leq k$, and with every N_n, $n \geq 1$.
- **Census safety counts.** Among the 7,561 classes of size at most 10, exactly 1,693 avoid being left hunting stag alone against every agent in that census, and 517 of these hunt stag with themselves; tested against all

opponents of size at most 12, only one of the 517, $\Box(p \vee (q \wedge \neg\Box^3 q))$, is exploited (by an opponent of size 11).

Why it matters

Safe agents in the Stag Hunt must refuse to hunt stag unless they can be sure of a partner, which pulls against coordinating with simple partners such as E_1 and N_1. The result locates the smallest size at which safety and these coordination properties can be combined, and gives an explicit infinite family that does it.

Proved vs checked

The laws for X_k, including safety against every modal opponent, are proved for all k in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [29], cited, not formalised (see also [12]). The non-existence below size 13 and the census counts are exhaustive computations up to the stated sizes, with every exploiting opponent recorded and re-solved.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire et al., arXiv 1401.5577.
- LessWrong / Alignment Forum wiki "Modal combat" and "Is there only one FairBot?" (2026).
- Mennen, "Modal combat for games other than the prisoner's dilemma" (2017).
- Todorov, ten Napel & Müller, "Parametric Open Source Games", arXiv 2606.27068.
- Tennenholtz (2004), program equilibrium (background).
- Web searches for modal agents in the Stag Hunt and for tournaments of modal agents (2026-10-21); none means not found in this search.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K9) and partly checked by script (claims K16, K13, K19); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K16

In guarded L of syntax-node size ≤ 12, exactly 356 of 56,503 classes get stag/stag with self, E_1, and N_1; every one is exploited by a size-≤ 10 opponent. Hence no universally safe agent of size ≤ 12 has all three coordination properties.

Status: derived ů author check only.

Verified.

- `checks/K16-no_safe12.py`: SHA256-pinned full enumerator; 56503 classes through size 12; independent machine and formula filters give exactly 356 self/E1/N1 candidates; each has re-solved size<=10 exploitation witness; 362 GLS certificates. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K9

For all $k \geq 1$, $X_k = \Box(p \vee \Box^k p) \wedge (\Box\bot \vee \neg\Box\neg p)$ never ends hunting stag against an eventual hare hunter, ends stag/stag with itself, FairBot and every R_m, D_m, X_m, with E_j $(j \geq 1)$ iff $j \leq k$ and with N_n iff $n \geq 1$.

Status: derived ů author check only.

Verified.

- `checks/K9-c046_patient_laws.py`: Lean 16 theorems (xk_safe, xk_self, xx/xr/xd_pair, xk_db, xk_E, xk_N, xk_Z, xk_modal,...), no sorryAx, false bound rejected; 452 height cells k<=7; 12 GLS certificates; census size<=10: 1693 safe, 517 safe self-stag, 230 SS with E_1, 75 with N_1, 0 with both — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Height semantics on the GL chain (box holds at h iff its argument holds at all g\<h); game tails equal standard-model actions, Lean proves laws for any traces solving the height equations.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K13

Among the 7561 guarded L classes of syntax-node size ≤ 10, exactly 1693 avoid stag/hare against every census opponent; exactly 517 of these hunt stag with themselves. These are finite-census safety counts.

Status: derived ů author check only.

Verified.

- `checks/K13-todorov_safety.py`: SHA256-pinned solver; all 7561-class cells re-solved; unique winner A3 at (3,0,2,1), score 16393 versus FairBot 14158; 7561 winner cells independently re-solved; 4 GLS certificates; 1693 census-safe classes, 517 self-stag. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K19

Of the 517 size-$\leq$ 10 census-safe self-stag agents, exactly one is exploited by a size-$\leq$ 12 opponent: $V = \Box(p \vee (q \wedge \neg\Box^3 q))$, by $B = \neg\Box^2 p \vee \Box(p \wedge \Box^2 q)$ of minimum size 11. The other 516 survive this finite test.

Status: derived ů author check only.

Verified.

- `checks/K19-safety12.py`: Pinned full enumeration: 56503 classes; 7561-class matrix recovers 1693/517; 25303014 extension cells at exact joint fixed states; exactly V refuted, 516 survive; minimum exploiter size11; 7561 independent V row cells; exploitation and self-play GLS certificates. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261021-0517`, claims K16, K9, K13, K19, run-folder commit `580565ffd3b3f1a8ca3ffa0407c949c8d724865c`, published 2026-10-21.

3 Discussion

The census counts are finite statements: "safe" there means safe against the agents in the census, and tests against larger opponents can still find exploits, as they did for one agent. The size-13 agent is safe against every modal opponent by proof. Safety is the same outcome condition as unexploitability in the prisoner's dilemma; what changes in the Stag Hunt is which coordination properties one wants alongside it.

4 How to verify

Each finding has its check scripts in `claims/staghunt-safe-coordinators/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.